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Long-term spatial memory across large spatial scales in *Heliconius* butterflies

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Locating food in heterogeneous environments is a core survival challenge. The distribution of resources shapes foraging strategies, imposing demands on perception, learning and memory, and associated brain structures. Indeed, selection for foraging efficiency is linked to brain expansion in diverse taxa, from primates[1] to Hymenopterans[2]. Among butterflies, *Heliconius* have a unique dietary adaptation, actively collecting and feeding on pollen, providing a source of essential amino acids as adults, negating reproductive senescence and facilitating an extended longevity[3]. Several lines of evidence suggest that *Heliconius* learn the spatial location of pollen resources within an individual's home range[4], and spatial learning may be more pronounced at these large spatial scales. However, experimental evidence of spatial learning in *Heliconius*, or any other butterfly, is so far absent. We therefore tested the ability of *Heliconius* to learn the spatial location of food rewards at three ecologically-relevant spatial scales, representing multiple flowers on a single plant, multiple plants within a locality, and multiple localities. *Heliconius* were able to learn spatial information at all three scales, consistent with this ability being an important component of their natural foraging behaviour.

Heliconius establish 'traplines', foraging routes along which specific plants are repeatedly visited with high spatial and temporal regularity, initiated from a stable roosting site[3]. Traplines are highly individualistic, even among butterflies that share a roost, suggesting that memory is guiding their movements[5]. When translocated hundreds of meters, wild *Heliconius* (*erato* and *melpomene*) also quickly orientate towards, and return to, their site of origin[6]. However, many assumptions about spatial learning in *Heliconius* remain untested.

To experimentally validate spatial learning in *Heliconius* across spatial scales, we performed three experiments at increasing spatial distances. In the first experiment, freshly eclosed, insectary-reared *Heliconius erato phyllis* (n = 44) were trained to locate rewarding feeders on a 100 cm^2 4 x 4 grid of 3 cm flowers (Figure 1A). We trained butterflies to associate either *border* (Figure 1A, column A or D) or *centre* (B2/3, C2/3) flowers with a positive food reward. After five days of training, we recorded individual preference using empty feeders. Comparing individual preferences before and after training, we found that trained butterflies showed an increased preference for the position of positively rewarded flowers (Figure 1D; $\chi^2_1 = 4.908$, $p = 0.027$), with no difference in performance between *border* and *centre* groups ($\chi^2_1 = 0.097$, $p = 0.756$).

We subsequently increased the spatial scale by conducting a similar experiment in a ~3 m^2 two-armed maze (Figure 1B). Individuals (n = 26) were released centrally, and we recorded their naïve preference for foraging in the left or right arm. The butterflies were then trained to associate one arm with a positive reward before a final preference test. We again found an effect of training on foraging preference (Figure 1E; $\chi^2_1 = 22.949$, $p < 0.0001$). There was a significant interaction between the training effect and direction of the reinforced feeder ($\chi^2_1 = 11.875$, $p < 0.001$), with butterflies trained to the left showing greater fidelity to the learned spatial cue (Figure 1E). Nevertheless, both *left* (n = 13, $\chi^2_1 = 11.639$, $p < 0.001$) and *right* (n = 13, $\chi^2_1 = 9.254$, $p = 0.002$) groups showed a significant learning effect.

Comparing the results of experiment one and two we found a significant interaction between the training effect and experimental design, with a stronger learning effect in experiment two (Figure 1D,E; $\chi^2_1 = 31.661$, $p < 0.0001$). This could be consistent with superior learning performance with more spatially distant resources. We therefore performed a third experiment that approaches the spatial scale over which wild *Heliconius* forage between pollen resources. Using the Metatron facility in Ariège, France[7], we created ~60 m wide T-mazes by connecting four cages

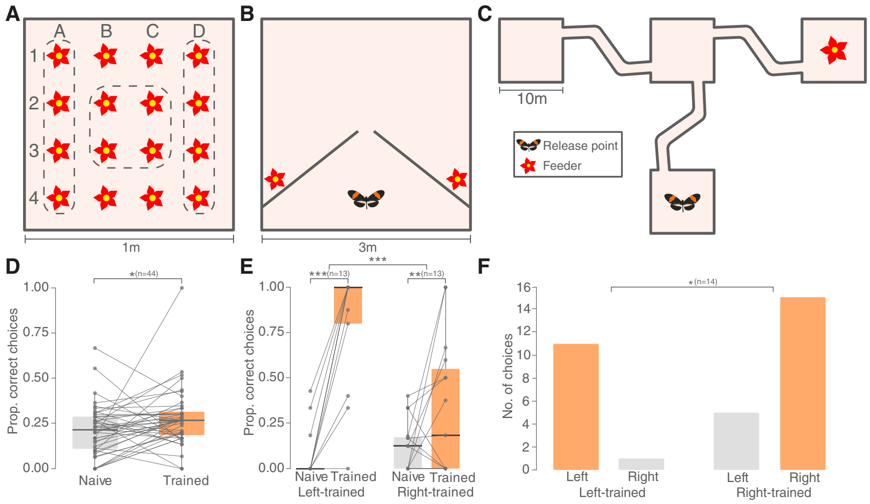

Figure 1 Spatial learning across different scales in *Heliconius*.
(A) 1 m² grid of 16 feeders, dashed lines indicate training groups: *border* group butterflies were trained to either column A or D flowers, *centre* group to B2, C2, B3 and C3. (B) 3 m² cage where butterflies were trained to associate food with either the left or right feeder. (C) T-maze composed of 10 m² cages. Butterflies were trained with a feeder in either the left or right arm. (D) Butterflies trained in (A) showed a small but significant shift in feeding preferences. (E) Butterflies trained in (B) showed a more substantial shift in preference, though accuracy was higher for butterflies trained to the left. (F) Butterflies trained in (C) showed a significant preference for foraging on the rewarded arm. ** = $P < 0.01$, *** = $P < 0.001$. ‘Correct choice’ indicates a feeding attempt on the feeder(s) the individual experienced as rewarding during training.

(Figure 1C). We trained *H. melpomene* to associate a food reward with either the left or right arm of the T-maze over 4 days. During training, we released butterflies in the base cage of the T-maze each morning (Figure 1C) to forage freely throughout the day. After training, butterflies were tested individually over the two days. In total, 32 trials were completed by the 14 *H. melpomene*. We found that individuals were more likely to enter the previously rewarded arm, consistent with a spatial memory of a food reward (Figure 1F, Table S1; $\chi^2_1 = 6.242$, $p = 0.012$).

Our results show that *Heliconius* can learn the location of food rewards at different spatial scales, and are adept at learning spatial cues over large scales, consistent with the apparent importance of long-range spatial learning for pollen feeding[4,5]. While the mechanisms by which *Heliconius* navigate remain unknown, in other insects the memorisation of visual scenes plays a primary role[8]. During our experiments, the sun and landscape features were partly visible through the cage walls and may have been used as a navigational aid. In common with adept spatial foragers such as ants and bees[2], *Heliconius* also exhibit markedly expanded mushroom bodies, which are the largest known in Lepidoptera[9]. This expansion is driven by increased visual input to the mushroom body, consistent with *Heliconius* relying on visual cues during navigation[9]. Our data constitute the first experimental evidence for spatial learning in Lepidoptera. Given the range of contexts in which butterflies are known to develop learnt preferences , spatial learning may be more common than appreciated. Nevertheless, the derived neural and behavioural traits of *Heliconius*, and their recent origin, offer a fruitful avenue for investigating the neural basis and evolution of visually guided navigation and enhanced spatial memory.

SUPPLEMENTAL INFORMATION

Supplemental information includes experimental procedures, one figure and one table, and can be found with this article online at https://doi.org/10.1016/j.cub.2023.06.009.

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AUTHOR CONTRIBUTIONS

All authors contributed to the experimental design. P.A.M. and F.J.Y. conducted the experiments and analyzed the data, under the supervision of M.Z.C. and S.H.M., and contributed equally. P.A.M., F.J.Y. and S.H.M. wrote the draft manuscript and all authors edited and approved the final version.

DECLARATION OF INTERESTS

The authors declare no competing interests

INCLUSION AND DIVERSITY STATEMENT

We support inclusive, diverse, and equitable conduct of research.

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